

Arbitrage-Constrained Neural Interpolation of the Implied Volatility Surface: Evidence from NSE Index Options

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ABSTRACT We study a neural implied-volatility surface whose total variance cannot decrease with maturity, by construction. Two earlier soft-penalty designs fail in one specific spot: near the money, at long maturities, where real quotes are sparse. There, training can flatten the maturity slope instead of fixing its sign, which lowers the penalty without solving the problem. Our hard-constrained model closes off that option. It builds total variance from increasing maturity basis functions with non-negative coefficients, so the slope cannot go negative no matter what the data looks like.

We test the model on 6.1 years of NIFTY 50 index options: 18 expanding-origin folds and 9,438 daily option-maturity slices, compared against five classical baselines. The static network has no measured calendar or butterfly violations. Raw SVI (stochastic-volatility-inspired) fits break calendar monotonicity on 60.0% of consecutive-expiry pairs. Thin-plate splines and cubic splines break the butterfly condition on 89.4% and 82.0% of slices, even inside the range of strikes actually observed.

Two baselines avoid both problems: surface SVI (SSVI), and Fengler's constrained call-price smoother, which we implement here as a quadratic program. Fengler's method is arbitrage-free in price space by construction. It fits every slice and reaches an in-range held-out RMSE of 0.45 volatility points, close to thin-plate splines' 0.42. So on this data, an arbitrage-free and accurate surface does not require a neural network. Giving the network two daily state inputs cuts its matched test-window error from 5.10 to 3.35 volatility points, while it still has zero calendar violations; SSVI still does better, at 2.18. We also withhold an entire interior 30–60 day maturity band from training. The network's accuracy on that withheld band barely changes, which is what its design predicts. But classical baselines given a simple monotone interpolation rule reconstruct the same band two to four times more accurately. The guarantee is real. Here, it does not pay off.

INDEX TERMS Implied volatility surface, no-arbitrage constraints, neural networks, constrained function approximation, SVI, SSVI, option pricing, NSE, NIFTY, walk-forward evaluation.

I. INTRODUCTION

The implied volatility (IV) surface maps strike and maturity to the Black–Scholes-equivalent volatility that reprices a traded option [1]. A useful surface should match market quotes without allowing butterfly or calendar arbitrage [2]. Ordinary cubic and thin-plate splines fit closely but give no shape guarantee. Shape-constrained spline methods can supply one [3], [4]. SVI and SSVI take a different route: low-dimensional parametric forms with explicit no-arbitrage conditions [5], [6].

Neural networks can enforce shape either softly or hard. A soft penalty, evaluated by automatic differentiation, can push

violations down [7]. A hard architectural constraint can make a condition hold for every input [8]. This paper asks what a hard construction actually buys on NIFTY 50 index options, under an expanding-origin evaluation, and why two earlier soft versions failed first.

This paper makes four contributions. It traces the soft calendar penalty's failure to a near-flat solution in a data-sparse region, and shows algebraically how a positive maturity-basis construction closes that escape route. It measures how often raw SVI, SSVI, thin-plate splines, and ordinary cubic splines violate the same dense-grid rule, and where: this is a measurement, not a correction, since the literature never claimed raw

SVI or unconstrained splines were arbitrage-free. It applies the construction to 6.1 years of NSE data, an emerging-market setting not previously studied with this architecture as far as we know. And it documents the real accuracy cost of fitting one static surface across a period where the volatility level and smile keep changing, then tests a state-conditioned extension that keeps the calendar proof intact.

Everything we find applies to our own hard and soft formulations. We do not implement the architecture of Zhang, Li, and Zhang [8], so we make no claim to have re-tested their model directly.

The rest of the paper runs as follows. Section II reviews parametric surface models and soft- and hard-constraint neural approaches, and places this paper against the closest prior work. Section III describes the NSE data, the parsing and filtering pipeline, and the frozen evaluation panel. Section IV describes the constrained architecture and the two soft-penalty attempts that failed before it. Section V describes the walk-forward evaluation protocol, including a pre-registration violation we disclose and how we handled it. Section VI reports fit accuracy, arbitrage-violation rates for every model, formal significance tests, error decompositions, robustness cuts, and wall-clock cost. Section VII covers what the results do and do not support. Section VIII concludes and lists limitations.

II. RELATED WORK

A. PARAMETRIC ARBITRAGE-FREE SURFACE MODELS

The stochastic volatility inspired (SVI) parameterization [5] models a single maturity slice's total implied variance $w(k) = \sigma^2(k)\tau$ as a function of log-moneyness k , using five free parameters. These were chosen to match the linear behavior that many stochastic volatility models predict for total variance in the wings. Gatheral and Jacquier [6] derive sufficient conditions on the five raw-SVI parameters for a fitted slice to be free of butterfly arbitrage. They also extend the parameterization to a full surface (SSVI), coupling an at-the-money total variance term structure $\theta(\tau)$ across maturities through a power-law term $\varphi(\theta)$. An isotonic (non-decreasing) $\theta(\tau)$, combined with a bound on $\theta\varphi(\theta)$, is then sufficient for the whole surface to be free of both butterfly and calendar arbitrage at once. These conditions underlie the SVI and SSVI baselines fit in this paper (Section III). We also use Durrleman's necessary and sufficient no-butterfly condition on total variance and its strike derivatives, $g(k) \geq 0$ [2], as the common arbitrage-scoring rule for every method in this study, parametric or not. Roper [9] gives general sufficient, and nearly necessary, conditions for a static-arbitrage-free implied-volatility surface.

B. SOFT-PENALTY NEURAL SURFACES

Ackerer, Tagasovska and Vatter [7] fit a feed-forward network to the IV surface with a training loss that penalizes violating no-arbitrage conditions, evaluated by automatic differentiation at collocation points sampled across the domain. This is a standard physics-informed neural network (PINN) approach,

and this paper's first architectural attempt (Section IV) follows it too. Horvath, Muguruza and Tomas [10] use neural networks differently: to learn the map from model parameters to implied volatility surfaces, for faster pricing and calibration in rough volatility models. That is model calibration rather than market-surface fitting, but we include it as a representative of the broader "deep learning for volatility" literature this paper's architecture also belongs to.

C. HARD-CONSTRAINT NEURAL AND NON-PARAMETRIC SURFACES

Arbitrage-free interpolation predates neural methods. Kahalé [3] builds a smooth arbitrage-free interpolation from discrete quotes. Fengler [4] formulates arbitrage-free smoothing of call prices with natural cubic splines and linear shape constraints, giving a quadratic program that is cheap to solve. Fengler's method is the standard non-parametric comparator for this problem, and we implement it as a fifth classical baseline on the same protocol (Section IV-A). It matters a great deal for how the neural result should be read: it is arbitrage-free by construction and about as accurate as the best unconstrained smoother in our panel. That removes "constrained and still reasonably accurate" from the list of properties that could set a neural surface apart here.

Zhang, Li and Zhang [8] propose a two-step framework in which a deep network's surface-reconstruction stage carries a hard architectural no-static-arbitrage constraint. They report that this hard constraint not only removes static arbitrage but also lowers prediction error relative to a standard unconstrained interpolation method on their data. Their finding motivates our comparison, on a different market and with a different construction. We do not reproduce their architecture, so our evidence bears only on the hard and soft formulations built here. Chataigner, Cousin, Crépey, Dixon and Gueye [11] take a more skeptical, complementary position: they build a provably arbitrage-free local-volatility surface with a shape-constrained Gaussian process instead of a neural network, and frame hard shape constraints explicitly as a trade-off against model flexibility and implementation complexity. That is a useful counterpoint, and we return to it when discussing SSVI's own empirically arbitrage-free behavior in Section VII. Cohen, Reisinger, and Wang [12] impose no-arbitrage restrictions in neural stochastic differential-equation market models, a dynamic setting that complements the static interpolation problem studied here.

Monotone-network constructions more broadly enforce derivative signs through positive integrands, constrained weights, adapted activation functions, or a fixed positive-combination structure like Equation 5. This lineage runs from early monotonic-network architectures [13] and their extension to partially-monotone inputs [14], an IEEE Transactions on Neural Networks result directly relevant to our own strike-conditioned, maturity-monotone construction, through calibrated lattice and look-up-table models [15], [16], general convex-network constructions [17], and recent scalable reformulations [18]–[21]. Equation 5 is a low-dimensional,

positive-basis member of this same family, specialized to a single monotone axis (maturity) rather than a general partially-monotone function.

A separate and more recent line of work targets arbitrage-free implied volatility surfaces through generative modelling rather than direct interpolation: variational autoencoders trained on the whole surface [22], simulation of surface dynamics under no-arbitrage constraints [23], and a GAN-based generator for daily surfaces [24]. Closer to this paper's setting, recent work also revisits deep-learning option pricing conditioned directly on the market's implied volatility surface [25]. These target simulation and pricing rather than the single-day interpolation and cross-date generalization problem studied here, but they confirm that shape-constrained, arbitrage-aware neural modelling of the surface remains an active area.

D. POSITIONING OF THIS PAPER

Relative to [8] and [11], this paper is not claiming to discover that a hard architectural constraint can enforce no-arbitrage in a neural surface. That is established prior work. What is new here is: (i) a documented, mechanistic account of *why* two different soft-penalty formulations failed on this data before we adopted the hard constraint; (ii) a direct measurement of how often classical baselines violate arbitrage conditions, under the same protocol used to certify the neural surface, which changes how the comparison should be read (Section VI, Section VII); (iii) the market itself, NSE index options, not previously studied in this literature as far as we know; and (iv) a simpler hard-constraint design, a strike-only network multiplying a fixed, non-negative basis of monotone maturity functions, rather than a constrained deep network [8] or a shape-constrained Gaussian process [11]. We treat (i) and (ii) as the paper's main contributions: each is a finding about constrained function approximation in general, independent of the exact market or architecture used to show it. We treat (iii) and (iv) as the application and the specific design that made the demonstration possible.

III. DATA

A. SOURCE AND SAMPLE PERIOD

The raw data is NSE's daily futures-and-options (F&O) bhavcopy, a per-trading-day file listing every listed contract's OHLC prices, settlement price, contract volume, and open interest, for **2020-01-01 through 2026-01-30** (1,496 trading days, 6.1 years), filtered to the NIFTY 50 index (NIFTY) options and futures series. Only 3 trading dates are missing across the entire span (2020-04-14, 2021-03-30, 2026-01-15), consistent with NSE holiday closures. NSE changed its bhavcopy file format (the "UDiFF" format) on 2024-07-08, renaming instrument codes (FUTIDX $\rightarrow$ IDF, OPTIDX $\rightarrow$ IDO). The cutover is clean and non-overlapping, and the parsing stage (Section III-C) handles both formats explicitly. We use each contract's *settlement* price rather than its closing price. On illiquid rows the two can diverge a lot: one zero-volume row, for example, was priced 3.4% apart between

TABLE 1. Filter waterfall, raw options-with-forward panel to the frozen evaluation panel.

Filter	Rows remaining	Slices remaining
F0: options w/ forward	2,944,060	29,135
F1: $\tau \in [7/365, 1.0]$ y	2,162,119	15,671
F2: volume ≥ 10 , OI > 0	714,131	12,370
F3: $k \in [-0.35, 0.25]$	705,570	12,348
F4: price $>$ intrinsic+tick	681,835	12,346
F5: OTM only	474,073	12,289
F6: static-arbitrage-free	318,122	12,289
F7: ≥ 8 strikes/slice	307,458	9,438

close and settlement. Settlement is also the theoretically correct price for options that may not have traded at the close.

B. FORWARD PRICE AND RISK-FREE RATE

The Black-76 forward price [1] needed for inversion comes from the matching NIFTY index future's settlement price, per (date, expiry) slice, where one exists: 29.6% of slices, mostly monthly expiries. For the remaining 70.4% of slices, mostly weekly expiries that do not trade as futures on the NSE index, we recover the forward instead from put-call parity at the strike minimizing $|C - P|$ within that slice. We prefer a strike where both legs are liquid (positive volume and open interest), and fall back to the unconstrained minimizer only when no liquid pair exists. The risk-free rate is proxied by the RBI repo rate, forward-filled to daily frequency, in place of the 91-day T-bill yield we originally wanted; no free daily T-bill series was available. This is a documented simplification, and, as is standard, it has only a second-order effect on inverted implied volatilities relative to the forward price and the observed premium.

C. FILTERING AND IV INVERSION

Table 1 shows the filter waterfall applied to the parsed, forward-matched panel before implied-volatility inversion. **F1** restricts maturity to $\tau \in [7/365, 1.0]$ years. **F2** requires volume ≥ 10 and open interest > 0 . We tightened this from a plain volume > 0 threshold after a data-quality check (Section III-E) found that volume > 0 alone did not remove clearly noisy quotes. **F3** restricts log-moneyness to $k \in [-0.35, 0.25]$. **F4** removes quotes priced below intrinsic value plus a tick. **F5** keeps only out-of-the-money options, with each strike represented once by its more liquid OTM side. **F6** drops the 155,951 rows that break within-expiry strike monotonicity or convexity bounds on raw option prices, checked before any model is fit; this filter does not test calendar arbitrage. **F7** requires at least 8 strikes per (date, expiry) slice, since a slice with fewer points leaves too few strikes to both fit on and score under the Task A 80/20 strike hold-out (Section V). Implied volatility is then inverted from Black-76 with a vectorized Newton-Raphson pass and a per-row Brent's-method fallback for rows that do not converge; in this run, every row converged under Newton-Raphson alone.

F6's rejection rate is high, 32.9%, so we checked whether the forward estimator was driving it. The rejection rate is

36.3% for rows using a matched futures forward, 29.0% for rows using a liquid put–call-parity forward, and 13.0% for the small illiquid-parity fallback group. That ordering does not support the idea that single-strike put–call parity is mechanically causing most F6 rejections. It does not, on its own, prove the estimator is optimal, so Section VI-J tests this directly with a multi-strike regression-based forward. It finds that the forward construction is not a material source of error.

D. FROZEN PANEL AND DESCRIPTIVE STATISTICS

The frozen evaluation panel (`panel.parquet`) contains **307,458 rows** across **9,438 (date, expiry) slices**, from 329 distinct expiries. Of these, 78 are the calendar month’s latest expiry ("monthly") and 251 are an earlier same-month expiry ("weekly"); this split comes directly from the expiry calendar, uses no fixed weekday assumption, and holds up across NSE’s several changes to the weekly-expiry weekday over the sample. Columns are: trade date, expiry, option type, strike, settle price, open interest, volume, time to maturity τ , forward price and its source, log-moneyness $k = \ln(\text{strike}/\text{forward})$, risk-free rate, and inverted implied volatility. Overall implied volatility has mean 17.7%, median 15.8%, interquartile range [12.8%, 20.4%], and a maximum of 126% in the March 2020 COVID crash window, matching known India VIX behavior. Near-the-money ($|k| < 0.03$) median IV by year ranges from a low of 10.8% in 2023, a well-documented low-volatility year for NIFTY, to a high of 18.9% in 2020. Every year-by-year level matches independently known NIFTY volatility history, with no sign of a pipeline artifact.

E. A DATA ANOMALY, INVESTIGATED AND RETAINED

Manual inspection of randomly sampled daily smiles turned up one genuine anomaly, at roughly a 1-in-10 sampled-slice rate: on 2024-03-13 (expiry 2024-05-30), three adjacent out-of-the-money call strikes were priced at implied vols of about 11.5–11.8%, sandwiched between neighboring strikes at about 15–16%. We checked every field in that slice (forward, maturity, moneyness) and found no pipeline error. This looks like a real settlement-price anomaly in NSE’s raw data for that day, most likely microstructure noise in thinly-traded weekly-expiry middle strikes. Tightening the volume floor from > 0 to ≥ 10 (Section III-C) removed some similarly noisy slices elsewhere, but not this one. We chose not to hand-remove individual anomalous slices, since that would bias the sample toward whatever an analyst subjectively judges "clean." Instead we keep this and similar noise in the frozen panel. A central motivation for arbitrage-constrained fitting (Section IV) is exactly that a constrained model should be robust to this kind of raw-quote noise, and Section VI reports where in (moneyness, maturity) space each method’s errors concentrate, which would show up any real anomaly clustering.

IV. METHOD

We work throughout in total implied variance $w(k, \tau) = \sigma(k, \tau)^2 \tau$, the natural coordinate for stating no-arbitrage con-

ditions [2], [6]. A surface is free of *butterfly* (strike-direction, fixed-maturity) arbitrage if the Durrleman function

$$g(k) = \left(1 - \frac{kw'}{2w}\right)^2 - \frac{w'^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2} \quad (1)$$

is non-negative for every k at fixed τ , where $w' = \partial w / \partial k$ and $w'' = \partial^2 w / \partial k^2$ [2]. A surface is free of *calendar* (maturity-direction) arbitrage if total variance is non-decreasing in maturity at fixed log-forward moneyness $k = \ln(K/F_\tau)$,

$$\frac{\partial w}{\partial \tau}(k, \tau) \geq 0 \quad \forall k, \tau > 0. \quad (2)$$

Every model in this study is scored against both conditions, on the same dense (k, τ) grid described in Section V.

A. BASELINES

We fit four classical baselines under Section V’s Task A protocol: 80% visible strikes, scored on the held-out 20%. **Raw SVI** [5] fits five parameters (a, b, ρ, m, σ) per (date, expiry) slice, $w(k) = a + b(\rho(k - m) + \sqrt{(k - m)^2 + \sigma^2})$, subject to positivity, parameter-domain, and wing-slope restrictions, including $b\sigma(1 + |\rho|) \leq 2$ [6], enforced with constrained SLSQP optimization. These restrictions do not by themselves guarantee $g(k) \geq 0$, so we still score raw SVI on the Durrleman condition, and we keep only converged, feasible optimizer results. **SSVI** [6] fits one surface per day: first an empirical at-the-money term structure $\theta(\tau)$ (closest-to-zero- k total variance per expiry), isototonically regressed to be non-decreasing in τ , a sufficient condition for Eq. 2; then a joint SLSQP fit of (ρ, η, γ) in the power-law term $\varphi(\theta) = \eta\theta^{-\gamma}$, subject to $\theta\varphi(\theta)(1 + |\rho|) \leq 4$, which is sufficient for no butterfly arbitrage at every maturity at once. **Thin-plate splines** fit one radial-basis-function interpolant per day, jointly across that day’s expiries, using `RBFIInterpolator` with a thin-plate kernel [26] and light smoothing; this method carries no arbitrage constraint at all. **Cubic splines** fit one ordinary cubic spline per (date, expiry) slice in k ; this method also carries no arbitrage constraint, and should not be confused with Fengler’s constrained call-price smoother [4]. Neither thin-plate nor cubic-spline fitting uses information beyond a single day.

Fengler’s constrained smoother [4] is our fifth baseline, and the only classical one that is arbitrage-free by construction rather than by parametric restriction. It smooths *call prices* rather than implied volatilities. We work in normalised coordinates $x = K/F_\tau$ and $c = C/(e^{-r\tau}F_\tau)$, so that $c(x) = N(d_1) - xN(d_2)$. Here x is just an affine rescaling of K within a slice, so convexity in x is convexity in K , and fixed x is fixed $k = \log(K/F_\tau)$, the same calendar convention used everywhere else in this paper. Since the panel is out-of-the-money only, we convert out-of-the-money puts to calls by put-call parity before fitting. Writing $g_i = c(x_i)$ on the visible strikes and γ for the interior second derivatives of a natural cubic spline, the fit solves the quadratic program

$$\min_{g, \gamma} \frac{1}{2}(g - y)^\top W(g - y) + \lambda \gamma^\top R \gamma \quad \text{s.t.} \quad Q^\top g = R \gamma, \quad \gamma \geq 0, \quad (3)$$

together with the endpoint slope bounds $-1 \leq c' \leq 0$, monotonicity $g_i \geq g_{i+1}$, and the price bounds $\max(0, 1 - x_i) \leq g_i \leq 1$. Convexity plus the two endpoint slope bounds together imply $-1 \leq c' \leq 0$ on the whole interval, so this constraint set holds between knots too, not just at them. Q and R are the usual tridiagonal spline matrices. We eliminate $\gamma = R^{-1}Q^\top g$ analytically and solve the resulting smaller, dense program with a dual active-set method, since carrying the equality explicitly leaves the system badly conditioned whenever adjacent strikes sit close together.

Two implementation choices affect the comparison directly. First, W is diagonal, with $w_i = 1/\max(\text{vega}_i, 0.02)^2$ at the observed implied volatility. This makes the price-space objective close to a least-squares fit in implied-volatility space, which is the space every baseline here is *scored* in; an unweighted price fit would silently put all of its accuracy near the money. Second, we choose the smoothing weight λ by nested cross-validation *inside the visible 80% of strikes*: a further random 20% of the visible strikes is withheld for this, never the Task A hold-out, because tuning λ on the evaluation set would hand this baseline an advantage no other method gets. The value we select is $\lambda = 10^{-8}$.

We impose calendar ordering by sweeping each day's expiries from the longest maturity to the shortest, constraining each curve to lie below the fitted next-longer one. Because the spline's value at any point is a linear function of g , we impose the ordering as linear inequalities on a shared dense grid, not only at the current slice's own knots; enforcing it at knots alone would leave two splines with different knot sets free to cross in between. We impose the ordering only where both maturities have quotes, since outside that shared range the longer curve is an extrapolation, and bounding by it would not be a claim the data supports.

B. FIRST ATTEMPT: A SOFT-PENALTY NEURAL SURFACE (V1)

Our first neural architecture is a standard physics-informed formulation in the tradition of [7]: a feed-forward network $w_\phi(k, \tau)$, with hidden layers [64, 64, 64] and softplus activations (twice-differentiable, which Eq. 1's second derivative needs, with a softplus output enforcing $w > 0$). It trains with a composite loss

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \left(\overline{\max(0, -g(k))^2} + \overline{\max(0, -\partial w / \partial \tau)^2} \right), \quad (4)$$

where the two penalty terms are mean-squared violations of Eqs. 1–2, evaluated by nested automatic differentiation on collocation points resampled every training step (the strike range padded 20% beyond what we observe; the maturity range spanning the observed tenors). We sweep λ over $\{0, 0.01, 0.1, 1, 10, 100\}$, with 3 random seeds per value, on one representative fold. The butterfly penalty works cleanly: even $\lambda = 0.01$ drives the mean post-hoc butterfly-violation rate on a dense evaluation grid down to 0.4%, and $\lambda \geq 1$ removes it in all 3 seeds. The calendar penalty does not. Violation rates show no steady improvement with λ , and seed-to-seed variance is huge at every $\lambda \geq 0.1$: at $\lambda = 10$, for

instance, three seeds under identical hyperparameters give violation rates of 0.885, 0.123, and 0.830. That looks more like a coin flip decided by random initialization than a controllable function of the penalty weight.

We traced the failure to one small, concentrated region: near-the-money, long-maturity strikes, where real training data is extremely sparse (as few as 19 data points across a 5.5-year training window with $|k| < 0.02$ and $\tau > 0.3$). A locally negative $\partial w / \partial \tau$ in that region only adds a small *squared* penalty next to the data loss, so the optimizer gets a weak gradient signal to fix it. Raising λ just scales up that same weak signal, without reliably escaping the local optimum it settles into. We also tried a second formulation, v2, which we did not carry forward: it replaced the squared- L^2 penalty with an L^1 hinge on the slope, $\max(0, \text{margin} - \partial w / \partial \tau)$, on the idea that L^1 gives a non-vanishing gradient for small violations. This made the problem *worse*, with a 91% violation rate at $\lambda = 1$, because the hinge can be driven toward zero just as easily by shrinking $|\partial w / \partial \tau|$ toward zero from either side, not only by making it positive. Concentrating collocation points in the sparse region did not fix this either: even at $\lambda = 100$, after extended training, 100% of the dense grid still violated the margin, and the logged penalty plateaued near its margin value regardless of λ or further steps. The network settles for a cheap, near-flat local solution that keeps the penalty bounded without ever making real progress toward a genuinely positive slope everywhere. Both formulations fail for the same underlying reason: in a data-sparse region the model is free to represent *any* $w(k, \tau)$, including one with a locally negative slope, and a soft penalty on the slope gives the optimizer an escape hatch, flattening the surface locally, that is cheaper to reach than the intended fix of making the slope correctly signed everywhere. Reshaping the penalty does not close this escape hatch, because the problem is architectural, not a property of the loss function.

C. THE FIX: A HARD ARCHITECTURAL CALENDAR CONSTRAINT (V3)

Rather than try a third penalty shape, our final architecture removes the escape hatch by construction. We parameterize total variance as

$$w(k, \tau) = \text{softplus}(b(k)) + \sum_{i=1}^3 \text{softplus}(c_i(k)) \varphi_i(\tau), \quad (5)$$

where $\varphi_1(\tau) = \tau$, $\varphi_2(\tau) = \sqrt{\tau}$, and $\varphi_3(\tau) = \log(1 + \tau)$ are three *fixed*, non-decreasing basis functions of maturity alone, and $b(k), c_1(k), c_2(k), c_3(k)$ are four outputs of one small feed-forward network that takes *strike alone* as input (hidden layers [32, 32], softplus activations). Because every φ_i is non-decreasing in τ , and every coefficient $\text{softplus}(c_i(k)) \geq 0$,

$$\frac{\partial w}{\partial \tau} = \sum_{i=1}^3 \text{softplus}(c_i(k)) \varphi'_i(\tau) \geq 0 \quad (6)$$

for every k and every $\tau > 0$, by construction. Eq. 2 then holds identically, with no training needed to enforce it and

no λ term needed for it either. Butterfly arbitrage is still handled as in v1: a soft squared- L^2 penalty on Eq. 1, evaluated by automatic differentiation on resampled collocation points, weighted by λ . So the v3 λ -sweep is a pure butterfly-versus-fit-error trade-off, since calendar arbitrage is already guaranteed at every λ .

This construction is deliberately simpler than the closest prior hard-constraint architectures. It is not a jointly-constrained deep network over (k, τ) [8], and not a shape-constrained Gaussian process [11], but a small strike-only network multiplying a fixed, analytically monotone maturity basis. We chose this design so the calendar constraint's correctness follows directly from Eq. 5's algebra, rather than depending on a more complex proof. The neural models are optimized with Adam [27] at learning rate 10^{-3} ; v3 uses batch size 8192 for the data term and 2048 for collocation points, and trains for 18,000 Adam steps per fold (Section V).

Section VI reports two side effects of this construction that we verified but did not design for. Even v3's $\lambda = 0$ baseline, with no butterfly penalty at all, has a butterfly-violation rate that is already near zero, because the smaller, strike-only sub-network produces smoother smile shapes more easily than v1's joint network does. And v3 improves Task A/B RMSE over v1 at low-to-moderate λ , though this improvement is confounded with the architecture change itself, a strike-only decomposition versus a joint network, so it is not a clean, isolated test of the constraint mechanism alone. We state this limitation explicitly in Section VIII.

D. DAILY-STATE-CONDITIONED EXTENSION (V4)

The static v3 surface has no date or market-state input, so it cannot represent daily changes in the volatility level or smile. Its accuracy gap therefore mixes two things together: the cost of the calendar constraint, and the cost of pooling across many years without a state input. We address this with a conditional extension. From each day's visible 80% strike sample, we pick the quote closest to $k = 0$ at each expiry and regress its at-the-money total variance on maturity. The fitted total variance at 30 days, s_0 , and the fitted slope, s_1 , form a two-dimensional daily state $s = (s_0, s_1)$.

The coefficient network now receives (k, s) rather than k alone:

$$w(k, \tau; s) = \text{softplus}(b(k, s)) + \sum_{i=1}^3 \text{softplus}(c_i(k, s)) \varphi_i(\tau). \quad (7)$$

Since s is constant across the maturities of a daily surface,

$$\frac{\partial w}{\partial \tau}(k, \tau; s) = \sum_{i=1}^3 \text{softplus}(c_i(k, s)) \varphi'_i(\tau) \geq 0. \quad (8)$$

The hard calendar constraint survives unchanged. Test-day states use only the visible strike subset, and we score accuracy on that day's fixed 20% held-out strikes. The extension uses the same hidden-layer widths, maturity basis, optimizer, butterfly-penalty weight, and 18,000-step training budget as v3, with no new hyperparameter search. So this comparison

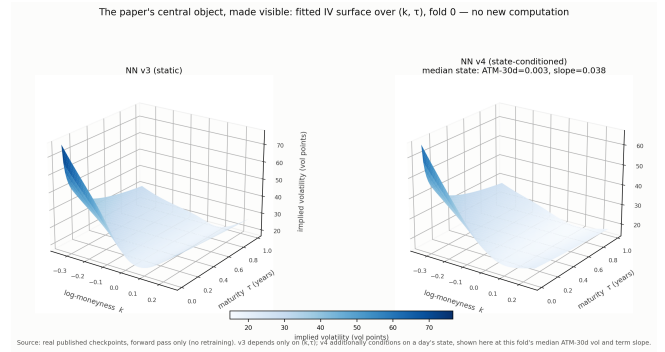


FIGURE 1. The published v3 and v4 checkpoints (fold 0) evaluated over a dense (k, τ) grid, forward pass only, no retraining. v4 is drawn at that fold's median daily state $s = (s_0, s_1)$, since a state-conditioned surface needs one state fixed to plot at all.

changes only the input conditioning, while every principal training setting stays fixed.

Fig. 1 shows both surfaces evaluated over a dense grid: the non-decreasing maturity slope Eq. 5 and Eq. 7 guarantee is visible directly, at every strike, rather than only in the aggregate violation tables of Section VI.

V. EXPERIMENTAL DESIGN

A. WALK-FORWARD FOLDS AND THE THREE EVALUATION TASKS

We use an expanding-origin walk-forward design: 18 folds, each with an 18-month/3-month/3-month train/validation/test split, stepping the origin forward 3 months per fold and spanning the panel's full date range (fold 17 is partial, with roughly one month of test data at the sample's end). Two tasks probe different kinds of generalization. **Task A** (interpolation) holds out a fixed random 20% of strikes per (date, expiry) slice, seed 42, the same across every method, and scores RMSE on those held-out strikes within the training window: how well a model fills in strikes it did not see, on dates it did see. **Task B** (cross-date evaluation) uses dates the network's weights never saw. For a fair comparison, every method is scored on the same fixed 20% held-out strikes on each test-window day. The classical baselines are refit on that day's visible 80%; v4 sees only the two state summaries computed from those visible quotes; v3 sees no test-day information at all. So Task B is a conditional same-day reconstruction test for the baselines and v4, and a genuine time-extrapolation test only for v3. We also keep v3's full-window score for continuity with the original design, but do not compare it directly with the held-out-only baseline scores.

Task C (maturity interpolation) probes an axis the other two cannot reach. Tasks A and B both hold out strikes and dates, but every quote they score is still at a maturity the model trained on, so neither speaks to how a model behaves between listed expiries. Task C removes an entire interior maturity band, 30 to 60 days to expiry, from the training data, and asks each method to reconstruct the withheld slices. We score only days that still have an expiry on both sides of the

band once it is removed, so this is interpolation in τ , never extrapolation past the ends of the term structure; 97.7% of days holding a band slice qualify. The per-slice and per-day classical methods only produce a smile at listed expiries, so we give each the strongest interpolation rule its structure allows: shape-preserving monotone cubic interpolation of $\theta(\tau)$ for SSVI, and of total variance at fixed k for SVI, Fengler, and cubic splines. That way the comparison measures the architecture, not the absence of a rule. Thin-plate splines need no rule, since they are already two-dimensional in (k, τ) . We score Task C in two windows: band slices dated inside the training window, where the network has seen the date and the daily-refit methods see the same day's remaining quotes, so the information available matches; and band slices in the test window, where the static network must also extrapolate in date.

Task A's held-out-strike scoring splits further into *in-range* (true interpolation, where the held-out strike falls inside that slice's visible strike range) and *out-of-range* (extrapolation, roughly 7% of held-out points, for example when the single highest or lowest listed strike ends up in the held-out set). We report both separately throughout, since naive cubic-spline extrapolation is known to behave badly outside its fitted range, and mixing the two would either hide that failure or unfairly penalize every method for a problem specific to cubic splines.

B. A DISCLOSED PRE-REGISTRATION VIOLATION

Architecture and λ selection looked at both tasks on fold 16, which our analysis plan did not allow. We therefore froze v3 at $\lambda = 0.01$, report fold 16 separately, and exclude it from every pooled test and robustness result. The other 17 folds have disjoint test windows and were not inspected during model selection; v4 inherits the frozen settings with no new search.

C. METRICS

We report errors in *vol points* throughout (implied volatility, in decimal, $\times 100$). Besides RMSE and MAE, Section VI also reports the median absolute error and a 10%-trimmed mean absolute error per method, to separate typical-day accuracy from tail-dominated behavior, motivated by cubic splines' heavy-tailed error distribution (Section VI). On per-date mean squared loss, we report a paired sign test and a Wilcoxon signed-rank test. These distribution-free tests keep the direction and frequency of daily wins visible even when rare spline explosions dominate mean squared error. We keep Diebold–Mariano statistics [28], with a Newey–West long-run-variance estimator [29], as a secondary check, with the caveat that this test was built for forecast comparisons rather than this mix of estimated interpolation and conditional reconstruction models.

D. ARBITRAGE SCORING

We score every method, the neural network and all four classical baselines, against Eqs. 1–2 on a common dense grid. For *butterfly*, at fixed maturity, we use 61 log-forward-moneyness

points spanning the visible range padded 20%, and report violations separately inside the observed range and in the padding. For *calendar*, on each day with 2 or more expiries, we compare each method's fitted $w(k, \tau_e)$ across consecutive expiries (sorted by maturity) on a shared per-day k grid, checking for non-decreasing total variance at fixed k . Because $k = \ln(K/F_\tau)$ and the forward varies by expiry, this is not a fixed-dollar-strike comparison; it is the same convention the SSVI calendar condition is built on. We compute the derivatives Eq. 1 needs in closed form for SVI and SSVI, using the exact algebraic derivatives of their functional forms; via *scipy*'s exact cubic-spline derivative for the cubic-spline baseline; via central finite differences (step 10^{-3}) for the thin-plate baseline, which has no closed form; and via automatic differentiation for the neural network. We validated every derivative implementation against finite-difference approximations on synthetic test cases before applying it to the real panel.

E. ROBUSTNESS AND UNCERTAINTY CHECKS

Two robustness cuts check whether the headline comparison (Section VI) is an artifact of one data subset: excluding the highest-volatility decile of trading days (using a constructed ATM, short/medium-maturity IV proxy, since the source data has no VIX-style index), and splitting by monthly versus weekly expiry cycle, derived from the expiry calendar itself rather than a fixed weekday assumption. A seed-variance check retrains the network from scratch with two more random seeds on every one of the 18 folds (54 checkpoints total, 3 seeds $\times$ 18 folds), to measure the noise floor any single-seed RMSE number in this paper should be read against. An earlier pass of this check ran only three representative folds, early, middle, and late in the walk-forward sequence; we report the full 18-fold version throughout, since it directly measures the noise floor at the same scale as the headline result rather than a 3-fold sample of it.

VI. RESULTS

A. THE HARD CONSTRAINT VERSUS TWO SOFT-PENALTY ATTEMPTS

Table 2 and Fig. 2 summarize the λ -sweep comparison between the soft-penalty architecture (v1, Section IV-B) and the hard-constrained architecture (v3, Section IV-C) on fold 16, 3 seeds per λ . This is the comparison our analysis plan calls the central methods-paper figure. Calendar-arbitrage violation is exactly 0.0 for v3 at every λ and every seed, against a range of 29–82% for v1, with no steady relationship to λ . Because v3's butterfly-violation rate is already close to zero even at $\lambda = 0$ (Section IV-C), the trade-off curve on the right-hand panel of Fig. 2 is nearly flat for v3. The intended design was to trade fit error against a shrinking but still-positive violation rate as λ falls; instead, v3 has very little butterfly arbitrage left to trade fit error for, at any λ . Task A/B RMSE is better for v3 than v1 at every $\lambda \leq 0.1$, and degrades less badly at high λ . Based on this sweep, we freeze $\lambda = 0.01$ as the recommended v3 configuration for every later result: it gives best-or-near-best

TABLE 2. v1 (soft penalty) vs. v3 (hard constraint), fold 16, 3-seed mean. Violation rates on a dense (k, τ) grid; RMSE is in decimal implied volatility (multiply by 100 for volatility points).

λ	Cal. v1	Cal. v3	RMSE v1	RMSE v3
0	0.367	0.000	0.0777	0.0650
0.01	0.303	0.000	0.0777	0.0613
0.1	0.294	0.000	0.0773	0.0648
1	0.824	0.000	0.1094	0.0843
10	0.613	0.000	0.1648	0.1562
100	0.590	0.000	0.2662	0.1490

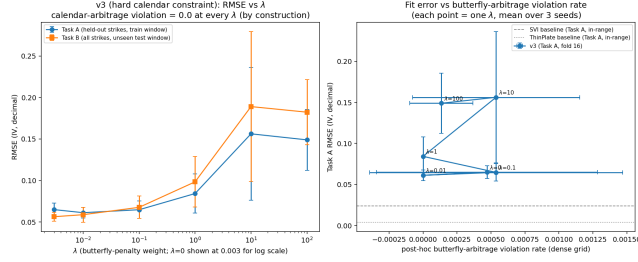


FIGURE 2. Fit error (RMSE) vs. arbitrage-violation rate, v3, across the λ sweep (fold 16, 3 seeds). Calendar-arbitrage violation is exactly zero at every λ by construction (Eq. 5); the butterfly-vs-fit trade-off (right panel) is nearly flat because v3's architecture already drives butterfly violation close to zero even without the penalty.

RMSE on both tasks, exactly zero calendar violation at any λ , and zero butterfly violation on the dense grid at this specific λ .

B. ISOLATING THE CONSTRAINT FROM THE DECOMPOSITION

Section VI-A's v1-versus-v3 comparison changes two things at once: the constraint mechanism (a soft penalty on $\partial w / \partial \tau$ versus the hard non-negative-coefficient construction of Eq. 5) and the functional decomposition (a joint network over (k, τ) versus a strike-only network multiplying a fixed monotone maturity basis). So v3's zero calendar violations cannot, from that comparison alone, be attributed to the constraint rather than to the decomposition. We isolate the two with a matched ablation that holds the decomposition fixed at v3's and varies only the constraint.

Two arms share every setting in Eq. 5 (layer widths, optimizer, batch sizes, the three fixed maturity basis functions φ_i , and an 18,000-step budget) and differ only in how the maturity coefficients $c_i(k)$ are produced. Arm D (*hard, calendar-only*) passes them through softplus, exactly as in Eq. 5. Arm E (*soft, calendar-only*) leaves them unconstrained and instead adds an L^1 -hinge penalty on $\partial w / \partial \tau$ with a strictly positive margin, the same penalty formulation used in v1/v2 (Section IV-B), so the soft arm fails, where it fails, at its strongest published form and not at a strawman. Both arms drop v3's butterfly penalty entirely: with unconstrained coefficients, w loses its positivity floor, and Durrleman's function (Eq. 1) contains $1/w$ terms that make the butterfly penalty explode for reasons unrelated to calendar monotonicity, which con-

TABLE 3. Hard (D) vs. soft (E) constraint, matched decomposition and butterfly-penalty-free, 3-seed mean $\pm$ sd, 17 blind folds.

Arm	Calendar viol. %	$w < 0$ grid %	Task B RMSE
D, hard	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	5.28 $\pm$ 0.51
E, soft	1.275 $\pm$ 0.053	2.087 $\pm$ 3.182	5.84 $\pm$ 0.86

founded an earlier version of this test that kept it. Because of this, neither arm is the published v3 model. A separate replication arm, run at the same settings as the published model with the butterfly penalty intact, matches its Task B RMSE (5.077 vs. 5.077) and its zero violation rate on all 18 folds to three decimal places, and is the bridge between this ablation and the model reported everywhere else in this paper.

Table 3 reports arms D and E across three seeds each on the 17 blind folds. Arm D has exactly zero calendar violation in every seed and every fold, 51 of 51 fold-seed combinations. Arm E violates calendar monotonicity in every seed, at a rate that itself spans two orders of magnitude across seeds: 0.039%, 0.141%, and 3.645% of the grid for seeds 1, 2, and 3. The same softplus construction that guarantees the calendar condition also guarantees $w \geq 0$ everywhere; this is a second, independent mechanism, foregrounded here rather than treated as a confound to route around. Arm E's grid fraction with $w < 0$, where implied volatility is not even defined, ranges from 0.01% to 5.75% across seeds (mean 2.087%), against exactly 0.000% for arm D in every seed and fold. A soft penalty can leave the surface undefined on part of the domain; a hard non-negative construction cannot.

On accuracy, arm D is no worse than arm E, not clearly better: mean Task B RMSE is 5.28 versus 5.84 volatility points, but the three-seed spread (0.51 and 0.86) is wide enough that the two are not statistically distinguishable, and one seed (seed 2: 5.12 vs. 5.21) has the soft arm slightly ahead. We state this as parity rather than an advantage: holding the decomposition fixed, the hard constraint buys an unconditional calendar-and-positivity guarantee without a measurable accuracy cost, not a measurable accuracy gain. The seed where arm E's calendar violation is worst (seed 3) is also the seed where its accuracy relative to arm D is worst (Task B RMSE 6.81 vs. 5.85): the soft arm's failures cluster together rather than trading off against fit quality.

A referee-style question is whether a larger penalty weight could rescue the soft arm. A supplementary sweep over $\lambda \in \{0, 0.01, 0.1, 1, 10, 100\}$, 3 seeds, found no weight that removes the violation entirely: at best 1 of 6 runs still violates, at a steep accuracy cost (training RMSE rising from 7.56 at $\lambda = 0.01$ to 57.22 at $\lambda = 100$). We report this only as suggestive rather than conclusive: the sweep ran in a different compute environment than the arms above, at a shorter training budget (4,500 vs. 18,000 steps) on only two folds, which were themselves selected because the soft arm had already failed there in an earlier single-seed pass. It is one more data point against a simple reweighting fix, not evidence that no λ works on the full panel.

TABLE 4. Neural dense-grid violation rate, mean across all 18 folds.

Model	Butterfly	Calendar
Static v3	0.0000%	0.0000%
State-conditioned v4	0.0043%	0.0000%

Read together, these results narrow rather than eliminate Section VI-A's confound. Most of v3's advantage over v1 (Table 2) comes from the decomposition, not the constraint alone: even the soft-penalty version of v3's own functional form (arm E) stays far below v1's 29–82% violation rate. But the constraint is what converts an empirical, seed-dependent near-zero into an unconditional guarantee, at no measurable cost to accuracy, and it is the only arm that also guarantees $w \geq 0$. We revise the limitations paragraph in Section VIII accordingly.

C. ARBITRAGE RESULTS AT FULL WALK-FORWARD SCALE

Table 4 separates the hard calendar result from the penalty-based butterfly result. Static v3 has no measured violations in any of the 18 folds. State-conditioned v4 also has zero calendar violations, with a strictly positive maturity derivative across the whole grid. Its added strike flexibility does produce a small residual butterfly failure in one fold: 0.0774% of that fold's grid points, or 0.0043% once averaged across all 18 folds.

Table 5 reports the same measurement for the four classical baselines across the 1,496-day panel. After we fixed the SVI acceptance bug, 8,862 of 9,438 slices (93.9%) had a converged, feasible fit; we omit failed fits rather than replace them with a non-converged result. SSVI has zero violations. Raw SVI breaks the butterfly condition on 4.0% of fitted slices and calendar monotonicity on 60.0% of fitted consecutive expiry pairs. Thin-plate and ordinary cubic splines break the butterfly condition on 90.8% and 84.3% of slices, and calendar monotonicity on 35.6% and 91.2% of pairs.

Separating the original 20% padding from the observed strike range does not change the spline result. Inside the observed range alone, 89.4% of thin-plate slices and 82.0% of cubic-spline slices still contain a butterfly violation. Padding widens the tested domain and adds further violations, especially for thin-plate splines, so we report both domains rather than pooling them into one headline number. For SVI, most of its remaining butterfly failures sit in the padding: 0.8% of slices violate within-range, against 3.9% in the padded region.

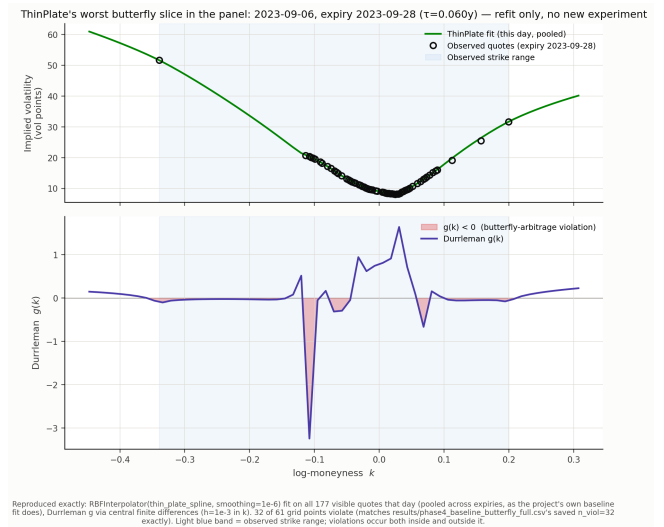
The butterfly sample holds 9,438 slices for SSVI and cubic splines, 9,436 for thin-plate splines, and 8,862 successful SVI fits; the calendar sample holds 7,942 pairs, 7,366 for SVI. The 9,504 count in the earlier draft came from accidentally concatenating a 15-day diagnostic chunk with the complete run. Key-based deduplication brings the result back in line with the frozen panel's 9,438 slices.

The architectural calendar guarantee sets both neural models apart from raw SVI and the unconstrained splines, but not from SSVI. Static v3's empirical butterfly result also matches

TABLE 5. Percentage of fitted slices or expiry pairs with at least one dense-grid violation. "Observed" uses the visible strike range; "padding" uses the 20% extension outside it. See the footnote below for the Fengler row.

Method	Butterfly all	Observed	Padding	Calendar pairs
SVI	4.0%	0.8%	3.9%	60.0%
SSVI	0.0%	0.0%	0.0%	0.0%
Fengler [†]	44.7%	3.8%	44.0%	1.7%
ThinPlate	90.8%	89.4%	68.9%	35.6%
CubicSpline	84.3%	82.0%	49.3%	91.2%

[†]Fengler's butterfly numbers above use the same w -space Durrleman test applied to every method, for comparability. Fengler's actual constraint acts on the call-price curve, where the guarantee is exact: zero convexity violations, zero slope-bound violations, and zero negative-density grid points across all 1,585 evaluated slices (Section VI-D). The w -space numbers here reflect the numerical stiffness of the price-to-implied-volatility coordinate change near the no-arbitrage boundary, not a failure of the constraint itself.

**FIGURE 3.** Durrleman's $g(k)$ (Eq. 1) for the thin-plate spline's single worst slice in the panel (2023-09-06, expiry 2023-09-28): 32 of 61 dense-grid points fall below zero (52.5%), matching the count already reported in Table 5. Shown here as a curve rather than a summary statistic, to make the shape of a butterfly violation concrete.

SSVI, and v4 shows why this result should not be called a hard guarantee.

D. THE CONSTRAINED NON-PARAMETRIC BENCHMARK

We report Fengler's smoother separately because its measurement is not quite like the other four. What the quadratic program actually constrains is the call-price curve, and there the guarantee is exact: across a sample of 250 days spread evenly over the 6.1 years (1,585 slices, 1,335 consecutive expiry pairs), there are *zero* convexity violations, *zero* slope-bound violations, and *zero* negative-density grid points. Since the fitted second derivative is the risk-neutral density up to a positive factor, this is the butterfly condition itself, not a proxy for it. The calendar sweep leaves 25 crossing grid points out of 24,549 inside the common support of consecutive maturities, or 1.7% of pairs containing at least one.

So the 44.7% figure in Table 5 is not a failure of the method,

but of the coordinate the comparison is made in. To score Fengler on the same Durrleman test as everything else, we have to convert the fitted price curve to implied volatility on the dense grid and differentiate it numerically, and near the no-arbitrage boundary that map is arbitrarily stiff. Restricted to the observed strike range, the rate is 3.8%, against 89.4% for thin-plate and 82.0% for cubic splines; almost all of the rest sits in the 20% padding, where the constrained price has been driven onto the intrinsic bound and no finite implied volatility exists at all (4.0% of grid points, reported as undefined rather than counted either way). We report the price-space numbers as the method's real arbitrage result, and the w -space numbers only for comparability.

On accuracy the method is strong. It fits all 9,438 slices, against 93.9% for SVI, and recovers a finite implied volatility at 99.99% of in-range held-out strikes. Pooled over all 1,496 days, its in-range Task A RMSE is 0.445 volatility points against thin-plate's 0.422, and out-of-range it is the best of every method tested, 2.31 against thin-plate's 2.43 and cubic spline's 446. Per fold (Tables 6 and 7) it is statistically indistinguishable from thin-plate splines on typical-day error, while carrying none of their arbitrage violations and none of cubic spline's tail.

This is the single most consequential number in the paper for how the neural result should be read, and it cuts against the network. A 2009 quadratic program, fitting a slice in milliseconds, already gives an arbitrage-free surface that is about as accurate as the best unconstrained smoother we tested. Whatever the case for a neural surface on this panel is, it cannot be that constraint and accuracy are hard to get together.

E. TASK A/B ACCURACY AT FULL WALK-FORWARD SCALE

Table 6 reports mean RMSE across the 17 blind folds (fold 16 excluded, per Section V-B) on matched random 20% strike hold-outs. Static v3 has lower test-window than training-window RMSE in 15 of 17 folds (mean 5.10 versus 6.68 volatility points); folds 6 and 7 reverse the ordering. Restricting the comparison to the same held-out composition leaves the mean gap close to the former full-window score of 5.08, but it does not support the earlier claim that the ordering holds in every fold. Fold 16 stays excluded from every aggregate.

State conditioning changes the size of the neural result. With the same architecture widths, penalty weight, and 18,000-step budget, v4 cuts training-window hold-out RMSE from 6.68 to 4.14 volatility points, and test-window hold-out RMSE from 5.10 to 3.35: reductions of 38.0% and 34.2%. V4 is still less accurate than SVI, SSVI, and thin-plate splines. In the test window, its gap to SSVI falls from 2.92 to 1.18 volatility points.

SVI, SSVI, and thin-plate splines beat the static network by roughly $1.7\text{--}10\times$ on both axes, and beat v4 too. Cubic spline's large mean RMSE (47–51) comes from rare, catastrophic per-fold blow-ups, one fold's test-window RMSE reached 480 vol points, a known instability of unconstrained cubic splines on sparse or irregular strike grids, and not a new finding of this

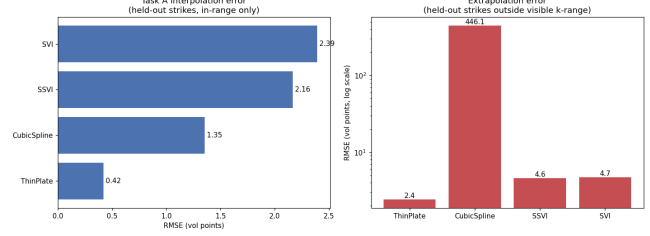


FIGURE 4. Classical baseline comparison, in-range vs. out-of-range Task A RMSE (vol points), all 1,496 days.

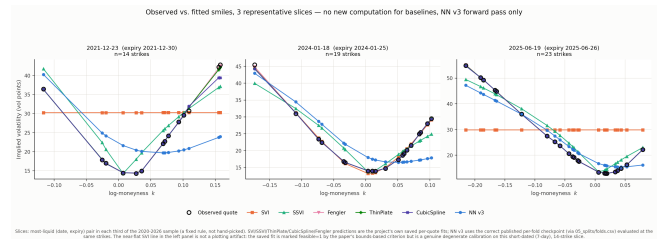


FIGURE 5. Observed vs. fitted smiles, all six methods, on three dates chosen by a fixed rule (most-liquid date/expiry in each third of the sample): 2021-12-23, 2024-01-18, 2025-06-19. Classical baselines use their own saved held-out predictions; NN v3 uses the published per-fold checkpoint at the same held-out strikes. The left panel's near-flat SVI fit is a genuine degenerate calibration on a short-dated, 14-strike slice, not a plotting artifact; it is one concrete instance of the calendar and near-the-money fragility Table 5 and Table 8 report in aggregate.

TABLE 6. RMSE (vol points), mean across the 17 blind walk-forward folds.

Method	Train hold-out	Test hold-out
NN v3, static ($\lambda = 0.01$)	6.68	5.10
NN v4, state ($\lambda = 0.01$)	4.14	3.35
ThinPlate	0.947	0.633
Fengler	0.833	0.646
SSVI	2.791	2.179
SVI	3.946	2.573
CubicSpline	47.59	51.09

evaluation. Fig. 4 shows this pattern in the original Task A in-range/out-of-range Phase 2 baseline comparison: the two unconstrained interpolators, thin-plate and cubic spline, fit visible market quotes much more tightly in-range than either arbitrage-constrained parametric model, SVI and SSVI. That is the same fit-versus-constraint tension the network is built to navigate, and cubic spline's extrapolation blow-up out-of-range is already visible before the network even enters the comparison.

SVI's accuracy entries condition on successful fits, so they cover 93.9% of slices; do not read them as full-coverage estimates. The other baselines cover essentially the full panel; Fengler's per-fold entries cover 99.0% of held-out strikes, the shortfall being out-of-range points whose constrained price sits on the no-arbitrage boundary. The NN v3 row is a single-seed figure; Section VI-G and Table 9 report the same methodology averaged over 3 seeds across all 18 folds (6.87 ± 0.32 / 5.21 ± 0.45), consistent with the single-seed headline within the seed-noise floor.

TABLE 7. Baseline error, test window, mean across 17 blind folds: RMSE vs. robust statistics (vol points).

Method	RMSE	MAE	Median err	Trim. mean err
ThinPlate	0.633	0.181	0.051	0.073
Fengler	0.646	0.203	0.055	0.080
CubicSpline	51.094	2.089	0.057	0.089
SVI	2.573	1.212	0.485	0.722
SSVI	2.179	1.469	1.006	1.189

TABLE 8. Paired daily mean-squared-loss tests, static NN v3 vs. each baseline (17 blind folds, $n=1,004$ days).

Baseline	Days baseline wins	Sign-test p	Wilcoxon p
SVI	838	$< 10^{-107}$	$< 10^{-107}$
SSVI	982	$< 10^{-257}$	$< 10^{-162}$
ThinPlate	1,003	$< 10^{-298}$	$< 10^{-165}$
CubicSpline	785	$< 10^{-74}$	$< 10^{-25}$

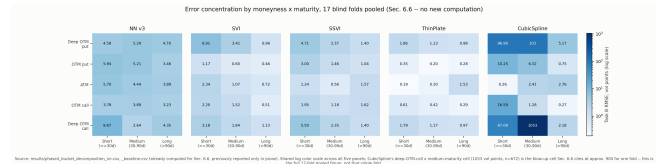
Table 7 adds median and 10%-trimmed-mean absolute error to the same 17-fold test-window sample, to separate typical-day accuracy from tail behavior. Cubic spline's median error (0.057) is essentially tied with thin-plate's (0.051), both far ahead of SVI and SSVI on a *typical* day, even though its RMSE is 25–80 $\times$ larger than every other baseline's. So cubic spline is not a globally unreliable interpolator; it is a locally excellent one with a heavy right tail (a handful of blow-up days per fold reaching RMSE in the hundreds).

F. PAIRED DAILY-LOSS TESTS

Table 8 reports distribution-free paired tests on per-date mean squared loss for the fixed held-out strikes. SVI, SSVI, and thin-plate splines win on 838, 982, and 1,003 of 1,004 blind days. Cubic splines win on 785 days despite their much worse aggregate RMSE; the sign and Wilcoxon tests recover the result a mean-based comparison hides. The secondary Diebold–Mariano statistic for cubic splines is -1.07 ($p = 0.286$, nominally favoring the network), because rare spline explosions dominate its mean and long-run variance. The DM statistics for SVI, SSVI, and thin-plate splines stay significant at $p < 0.0001$.

G. ERROR DECOMPOSITION, ROBUSTNESS CUTS, AND SEED VARIANCE

We decompose error by moneyness bucket (deep OTM put, OTM put, ATM, OTM call, deep OTM call) and maturity bucket (short $\leq 30d$, medium 30–90d, long $> 90d$). Both the network and every baseline show the expected pattern: worst in the deep-OTM buckets, best at the money. The network's degradation in the deep-OTM/long-maturity cells is generally milder in *relative* terms than the baselines' occasional blow-ups there (for example, cubic spline's deep-OTM-call $\times$ medium-maturity cell alone reaches RMSE ≈ 900 vol points in the fold we examined), but the network's *absolute* error level stays higher than thin-plate/SSVI/SVI in every bucket.

**FIGURE 6. Task B RMSE by moneyness $\times$ maturity bucket, five small-multiple panels (NN v3, SVI, SSVI, ThinPlate, CubicSpline), pooled across the 17 blind folds. CubicSpline's deep-OTM-call $\times$ medium-maturity cell reaches 1,053 vol points ($n=672$), the pooled version of the single-fold ≈ 900 -vol-point example already cited above.****TABLE 9. Seed-variance check: Task A/B RMSE (vol points), mean across the 17 blind folds, by seed (54 checkpoints, 3 seeds $\times$ 18 folds).**

Task	Seed 1	Seed 2	Seed 3	Mean $\pm$ sd
A, train hold-out	6.680	6.681	7.237	6.87 ± 0.32
B, test hold-out	5.077	4.835	5.706	5.21 ± 0.45

Two robustness cuts on the blind-pooled Task B sample confirm the headline comparison is not an artifact of a narrow subset of days. Excluding the highest-volatility decile of trading days moves every method's RMSE only a little (network 5.14 $\rightarrow$ 5.08 vol points). Splitting by monthly vs. weekly expiry shows the network less accurate on weekly expiries (5.38 vs. 4.93 vol points), while cubic spline's instability is far worse on monthly expiries (RMSE 181) than weekly (22), consistent with monthly contracts having sparser same-day strike grids at far tenors.

A seed-variance check retrains v3 from scratch with two additional seeds on every one of the 18 folds (54 checkpoints: 3 seeds $\times$ 18 folds), extending an earlier 3-fold pass of the same check. Table 9 reports mean RMSE across the 17 blind folds, computed per seed and then averaged. Seed 1 reproduces the published Task A headline exactly (6.680 vs. 6.68) and is close to the published Task B figure (5.077, before the matched-composition adjustment of Section VI-E gives 5.10); seeds 2 and 3 move the mean to 6.87 ± 0.32 on Task A and 5.21 ± 0.45 on Task B. The within-fold, across-seed standard deviation, averaged over all 18 folds, is 0.353 vol points on Task A and 0.508 on Task B, matching the original 3-fold sample's disclosed 0.34/0.54 almost exactly, now backed by 54 fold-seed checkpoints instead of 9. Fold-to-fold variation is larger than seed variation by a factor of about 2.2–2.4 (across-fold standard deviation at a fixed seed: 0.776 on Task A, 1.232 on Task B), so folds genuinely differ in difficulty and are not merely noisy, but seed noise alone is large enough to move a single-seed headline number by several tenths of a vol point. That is smaller than the gap to the classical baselines in Table 6, but not negligible when comparing configurations within the network's own λ -sweep (Table 2) or the isolating ablation (Section VI-B).

The matched 20% re-score in Section VI-E directly addresses the Task A/B composition confound: v3's mean test-window RMSE is 5.10 rather than 5.08, and stays below its 6.68 training-window RMSE, though only in 15 of 17 folds. A

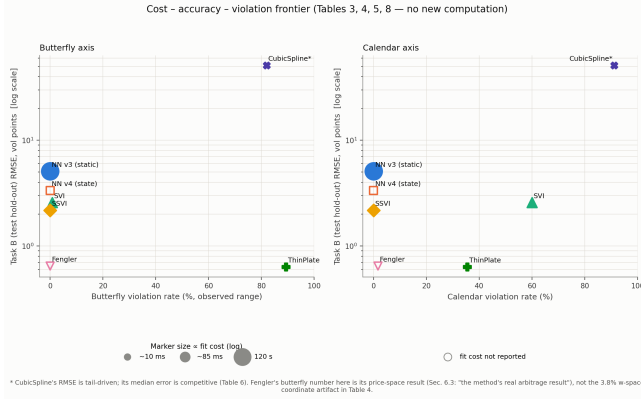


FIGURE 7. Cost-accuracy-violation frontier: Task B RMSE (log scale) against butterfly violation rate (left panel) and calendar violation rate (right panel), marker size encoding fit cost (Table 10). Fencler's price-space butterfly result (zero) is plotted rather than its w -space number, per the footnote to Table 5. Fencler and NN v4 use open markers at a fixed size, since neither has a reported fit cost in this paper; a filled marker sized by a number that does not exist would misrepresent the comparison. Fencler and SSVI sit in the low-error, low-violation corner of both panels; NN v3/v4 sit at zero-to-near-zero violation with visibly higher RMSE and, for v3, by far the largest fit cost of any method with a reported one.

TABLE 10. Wall-clock cost per unit of work.

Method	Fit	Predict
NN v3	120.1 s / fold (18k steps)	0.83 s / fold (all tasks)
SVI	84.9 ms / slice	included
SSVI	37.7 ms / day	included
ThinPlate	16.1 ms / day	included
CubicSpline	10.2 ms / day	included

secondary volatility-regime check gives $r = -0.82$ between the fold-level volatility gap and RMSE gap. This correlation is descriptive, and does not substitute for the matched-sample result.

H. WALL-CLOCK COST

Table 10 reports fit and prediction cost. The network's much higher fit cost does not buy any amortized-cost advantage at this scale. SSVI needs about $37.7 \text{ ms} \times 1,496 = 56.4 \text{ s}$ to refit the complete panel. The network needs about $120.1 \text{ s} \times 18 = 2,162 \text{ s}$ across the walk-forward folds. For a typical 63-day test window, SSVI refitting costs about 2.4 s against 120.1 s for one network fit. So the experiment supports no computational advantage over SSVI. The architectural difference is that the network defines a smooth, differentiable function of τ at arbitrary maturities, while the implemented SSVI only gets θ at that day's listed expiries and needs a separate interpolation rule between them.

I. TASK C: INTERPOLATING A WITHHELD MATURITY BAND

Tasks A and B both score at maturities the model was trained on, so neither one speaks to the claim that the surface works between listed expiries. Task C removes the interior [30, 60]-day band from the training data entirely, and asks each method

TABLE 11. Task C, reconstructing the withheld [30, 60]-day maturity band. RMSE and MAE in volatility points, mean across 17 folds (fold 16 excluded). c1 scores band slices dated inside the training window, where all methods have matched same-day information; c2 scores the test window, where the static network must also extrapolate in date.

Method	c1 train window		c2 test window	
	RMSE	MAE	RMSE	MAE
NN v3, band withheld	4.65	3.59	4.42	4.10
same net, band in training	4.62	3.58	4.42	4.12
Fencler + PCHIP	1.47	0.91	1.21	0.71
SSVI + PCHIP	2.11	1.49	1.73	1.24
SVI + PCHIP	2.63	1.61	1.71	1.04
ThinPlate (native 2-D)	2.51	1.58	1.65	1.02
CubicSpline + PCHIP	315	24.69	609	43.38

to reconstruct it. We score only days that still carry an expiry shorter than 30 days and one longer than 60 after the band is removed, so this is interpolation in τ for every method, never extrapolation off the end of the term structure; 97.7% of days holding a band slice qualify.

This comparison is only informative if the classical methods get the interpolation rule they lack, rather than being penalised for lacking it. So each is equipped with the best rule its own structure allows, using shape-preserving monotone cubic (PCHIP) interpolation so the rule cannot introduce a calendar violation of its own: for SSVI, interpolation of the ATM total-variance term structure $\theta(\tau)$, which is exactly the construction Section VII already conceded would close the functional gap; for SVI, Fencler, and cubic splines, interpolation of total variance across τ at fixed k . Thin-plate splines need no rule, since they are already a two-dimensional smoother in (k, τ) .

Table 11 also carries the control that makes the experiment readable: the published fold checkpoints, trained *with* the band present, evaluated on the same band quotes. The difference between the two neural rows is the cost of the maturity hole and nothing else. Comparing the withheld network only against the classical baselines would mix that cost together with the static model's overall accuracy level, which is a separate matter we already report elsewhere.

Two results follow, and they point in opposite directions.

The structural claim survives, cleanly. Removing an entire interior maturity band from training leaves the network's accuracy on that band statistically unchanged: the two neural rows differ by well under the seed-to-seed spread already reported in Section VI-G. This is exactly what the parametrisation predicts: Eq. 5 spans τ with three fixed monotone basis functions and non-negative coefficients, so a gap in the observed maturities removes almost no information about the shape in between. Nobody had shown this before, and it is the one place in this paper where the architecture does exactly what it was built to do.

The predictive claim does not survive. Every classical baseline given a monotone interpolation rule reconstructs the withheld band more accurately than the network does, by factors of roughly two to four; Fencler's constrained smoother is the most accurate of them in both windows, just as it

was in Task A. Being continuous in τ by construction is a genuine structural property, but it does not turn into a predictive advantage on this panel. Cubic splines are the exception that proves the rule, and not in a useful direction: their per-day median error is competitive, while their RMSE runs to hundreds of volatility points, the same heavy-tailed failure mode Section VI-E documents.

The honest summary is that the interpolation rule the classical methods needed turned out to be cheap: supplying it costs a few lines of code. A practitioner who wants an off-expiry smile on data like this panel should fit Fengler's smoother and interpolate its total variance, not train a network.

J. SENSITIVITY TO THE FORWARD CONSTRUCTION

The forward for each (date, expiry) comes from a matched futures settlement where one exists, and otherwise from a single-strike put-call-parity estimate at the nearest-to-the-money strike. Because every quantity in this paper is expressed in forward moneyness, an error in the forward propagates into k , into the implied volatility, and into the filters. So we re-estimate the forward from *all* liquid parity pairs in a slice. Put-call parity is exactly linear in the strike,

$$C(K) - P(K) = e^{-r\tau} (F - K), \quad (9)$$

so regressing the call-put spread on strike, over every strike where both legs traded, returns the forward and, as a free diagnostic, the implied discount factor. Restricting to $|k| \leq 0.10$ and at least four liquid pairs leaves 11,735 slices, with a median of nine pairs each.

The two estimators agree closely. The median absolute relative difference between the multi-strike forward and the Phase-1 forward is 3.8 basis points; the median signed difference is -0.15 bp for futures-based forwards and -0.14 bp for parity-based ones. The unconstrained regression's implied discount factor sits at 1.001 times the known one, so parity holds in these settlement prices to about ten basis points.

Rebuilding the entire derived panel on the multi-strike forward, rerunning filters F1–F7 and the Black-76 inversion end to end, moves it from 307,458 quotes and 9,438 slices to 308,048 quotes and 9,432 slices. The F6 static-arbitrage rejection rate, which the review asked about specifically, moves from 34.651% to 34.506%. On the 307,037 quotes common to both panels, the median implied-volatility shift is $+0.0001$ volatility points, the mean absolute shift is 0.121, and the 5th and 95th percentiles are -0.267 and $+0.279$. For scale, the accuracy differences this paper is arguing about run one to five volatility points, an order of magnitude larger. So the single-strike parity estimator is not a material source of error here, and the earlier finding stands: F6 rejection is highest for futures-implied forwards, not for parity-implied ones.

K. WHAT IS GUARANTEED AND WHAT IS MEASURED

Table 12 separates the two kinds of claim this paper makes, because mixing them together is the easiest way to overstate the result. A guarantee holds by construction for every input,

TABLE 12. Guaranteed by construction versus measured on this panel. "Guaranteed" means the property follows from the parametrisation or the constraint set for any input, not that it was observed to hold here.

Property	NN v3 / v4	Fengler	SSVI
Calendar monotonicity	Guaranteed	Constrained	Guaranteed
Butterfly / density ≥ 0	Empirical	Guaranteed	Guaranteed
Defined at unlisted τ	Guaranteed	Needs rule	Needs rule
Differentiable in (k, τ)	Guaranteed	In k only	In k only
Accuracy	Empirical	Empirical	Empirical
Full-panel coverage	Empirical	Empirical	Empirical

and can be checked just by reading the parametrisation. An empirical property holds on this panel, was measured, and could fail on another one.

The row that carries the paper's one remaining positive claim is the third. Both neural models define $w(k, \tau)$ in closed form at every $\tau > 0$, so a query between listed expiries needs no extra machinery. SSVI only gets $\theta(\tau)$ at listed expiries; Fengler produces one constrained curve per listed maturity, tied to its neighbours by the calendar sweep but not continuous in τ . Both therefore need an interpolation rule the network does not. Section VI-I tests whether that structural difference actually produces a measurable advantage once the classical methods are given such a rule.

VII. DISCUSSION

A. WHAT IS GUARANTEED

The neural construction guarantees calendar monotonicity: for a fixed daily state, total variance cannot decrease with maturity. That holds pointwise, and does not depend on collocation density or penalty tuning. Butterfly arbitrage is different. It stays controlled only by a soft Durrleman penalty. Static v3 happens to reach zero measured butterfly violations in every fold, but that is an empirical result, not an architectural theorem. The state-conditioned experiment makes this distinction visible, since its extra flexibility can improve fit while also bringing back small strike-direction violations.

The guarantee does not set the network apart from SSVI on this panel. SSVI records zero violations under the same grid test, and is more accurate. It is also cheaper under the protocol we actually ran: daily SSVI fits take 56.4 seconds over the complete panel, while the 18 neural fits take about 2,162 seconds. So the evidence supports neither an accuracy advantage nor a computational advantage over SSVI.

Fengler's constrained smoother narrows the space further, and in a more uncomfortable direction. SSVI gets its no-arbitrage properties by restricting the surface to a three-parameter family, so you could argue its constraint was bought with flexibility. Fengler does not make that trade: it is non-parametric, arbitrage-free by construction in price space, fits every slice in the panel, and its accuracy is statistically indistinguishable from the best unconstrained smoother we tested (Section VI-D). A reader looking for an arbitrage-free surface that also fits this market well does not need a neural network, and did not need one in 2009. This is the most

important finding of the evaluation: a negative result about the motivation for the architecture, not about its correctness.

What survives is narrower, and structural. Equation 7 defines a smooth, differentiable function of maturity in closed form, queryable at any $\tau > 0$. SSVI only estimates its ATM total-variance term structure at listed expiries; Fengler returns one constrained curve per listed maturity, tied to its neighbours by the calendar sweep but not continuous in between. Both therefore need an interpolation rule the network does not. The obvious objection is that such a rule is cheap to add: a shape-preserving interpolation of $\theta(\tau)$ would give SSVI much the same functionality. So the real question is not whether the network is the only object that can answer an off-expiry query, but whether being continuous in τ by construction buys any accuracy once the alternatives actually get their rule. Section VI-I runs that experiment, rather than leaving the claim resting on structure alone.

B. STATE CONDITIONING CHANGES THE ACCURACY RESULT

Static v3 has to represent 6.1 years of changing volatility levels and smiles with one function of (k, τ) . Its error therefore mixes two effects together: the restriction the maturity construction imposes, and the lack of a daily state. The matched v4 comparison separates part of this. Giving the coefficient network two summaries, estimated from the visible test-day quotes, cuts test-window error substantially while leaving the calendar proof unchanged (Table 6). So the static v3-versus-baseline gap cannot be read as the cost of a hard calendar constraint.

The remaining gap to SSVI should not be pinned entirely on the constraint either. The two state variables only capture level and a linear ATM term-structure slope, not smile skew, curvature, liquidity, or other daily variation. On top of that, v4 and the baselines perform conditional same-day reconstruction, while static v3 alone extrapolates with no test-day information. So this comparison identifies the missing state as a first-order source of error; it does not isolate a pure causal effect of the architectural constraint.

C. WHAT THE BASELINE MEASUREMENT ADDS

The classical-baseline result is a measurement, not a correction, of the literature. Raw SVI works slice by slice with no cross-maturity coupling, and ordinary thin-plate and cubic splines carry no arbitrage guarantee at all, so their violations are expected. What this paper adds is the scale and location of those violations under one shared rule: raw SVI has a small residual butterfly rate but a large calendar rate, and both unconstrained splines frequently break the butterfly condition even inside the observed strike range (Table 5). The in-range/padding split shows spline extrapolation is not the whole story.

This comparison would be incomplete without a constrained non-parametric method. Fengler's call-price smoother [4] is the obvious missing baseline, since it can combine spline flexibility with static-arbitrage constraints.

TABLE 13. Method choice by requirement, synthesized from the tables cited in the last column.

Requirement	Use	Source
Unconditional guarantee, both butterfly and calendar	SSVI	Table 12
Sub-second fit	ThinPlate/CubicSpline/SVI (10–85 ms)	Table 12
Off-expiry query, no added machinery	NN v3/v4, or ThinPlate (native 2-D)	Tables 12, 11
Best raw accuracy	ThinPlate (0.633), statistically tied with Fengler (0.646)	Table 6
Best accuracy with a guarantee attached	Fengler (price-space guarantee) or SSVI (full guarantee)	Table 6, §VI-D
Best typical-day error if tail risk is acceptable	CubicSpline (median err 0.057; RMSE 51 from rare blow-ups)	Table 7
Reconstruct an unlisted maturity band	Fengler + PCHIP, beats every method including the network by 2–4×	Table 11

Leaving it out would have limited any claim that a neural construction is needed to get both smoothness and no-arbitrage at once.

D. IMPLICATIONS FOR METHOD CHOICE

For same-day interpolation on data like this panel, SSVI gives the strongest combination we observed of accuracy, measured arbitrage control, and fit cost. Thin-plate splines give the smallest typical errors when violations can be tolerated or fixed downstream. The neural construction matters when a differentiable off-expiry surface and an algebraic calendar guarantee are actual requirements, but its butterfly behavior and accuracy still need validating. The most solid neural finding here is the mechanism behind the failed soft calendar penalties: sparse regions let optimization settle for a nearly flat slope instead of enforcing the right sign. Building the sign into the function class removes that failure mode.

Table 13 restates this as a practical selection guide, synthesized from Tables 6, 10, 11, and 12 rather than from any new measurement. No single method wins every row: the network's only unshared advantage is architectural continuity in τ , and Section VI-I's maturity-holdout test shows that continuity does not yet translate into the best off-expiry prediction on this panel.

VIII. CONCLUSION AND LIMITATIONS

This study isolates one specific benefit of a hard neural constraint. Two soft calendar penalties fail in a sparse region, because optimization can settle for a nearly flat maturity slope instead of making its sign positive. Representing total variance with non-negative coefficients on increasing maturity functions closes off that route and guarantees calendar monotonicity. On 9,438 NIFTY 50 option-maturity slices, static v3 has no measured calendar or butterfly violations

across 18 folds.

The baseline comparison narrows this claim, then closes most of it. Corrected raw-SVI fits break calendar monotonicity on 60.0% of fitted consecutive-expiry pairs; ordinary thin-plate and cubic splines frequently break the butterfly condition even inside the observed strike range. These numbers quantify known limitations of unconstrained methods. But two baselines do not share this limitation. SSVI records no violations, beats both neural models on accuracy, and costs far less under the refit protocol we tested. Fengler's constrained smoother, implemented here as a quadratic program on normalised call prices, is arbitrage-free in price space by construction, without restricting the surface to a parametric family: it fits every slice in the panel, is the most accurate method we tested out-of-range, and is statistically indistinguishable from thin-plate splines in-range while carrying none of their violations. Whatever case exists for a neural surface on this market, it cannot be that constraint and accuracy are hard to get together.

The one distinction that survived, a smooth, differentiable maturity representation queryable between listed expiries, is now tested rather than assumed. Withholding the entire interior 30–60 day maturity band from training leaves the network's accuracy on that band statistically unchanged. That is what the parametrisation implies, and no earlier experiment in this paper had actually shown it. But the same test also shows the property does not pay off: classical baselines given a shape-preserving monotone interpolation rule, including the $\theta(\tau)$ interpolation that would give SSVI the same functionality, reconstruct the withheld band roughly two to four times more accurately than the network does. So the structural advantage is real, and the interpolation rule that captures it is cheap.

Daily state accounts for a large part of the static network's error. Adding ATM total variance and its term-structure slope to the coefficient network cuts matched test-window RMSE from 5.10 to 3.35 volatility points, while keeping zero calendar violations. It does not close the gap to SSVI's 2.18, and one v4 fold keeps a small butterfly violation. So the static v3 accuracy gap is not just the cost of the hard constraint; it also reflects missing market state. Richer state conditioning is the clearest way to test how close a constrained surface can get to daily-refit accuracy.

A few limitations bound these conclusions. We inspected fold 16 during model selection, so we exclude it from headline aggregates; the remaining folds use one seed each for the headline tables, though a full 3-seed, 18-fold check (Section VI-G) puts the resulting noise floor at about ± 0.35 vol points on Task A and ± 0.51 on Task B. The v1-versus-v3 comparison in Section VI-A changes both the constraint mechanism and the functional decomposition at once; this is no longer an open confound. Section VI-B holds the decomposition fixed and varies only the constraint, and finds the hard construction adds an unconditional calendar-and- $w \geq 0$ guarantee at no measurable accuracy cost relative to a matched soft-penalty variant, while most of v3's overall

advantage over v1 still comes from the decomposition rather than the constraint alone. That ablation itself has narrower scope than the main results: its two arms omit the butterfly penalty and so are not the published v3 model, its accuracy comparison rests on only three seeds, and its supporting λ -sweep ran in a different environment, at a shorter budget, on two outcome-selected folds, so we treat that sweep as suggestive rather than conclusive. V4 uses only two state summaries from visible same-day quotes, so it is a conditional reconstruction model, not an unconditioned forecast. The implemented SSVI uses a two-stage ATM term-structure estimate rather than a joint fit. Fengler's smoother is constrained exactly in price space, but scoring it on the same implied-volatility grid test as the other methods needs a numerical conversion that is ill-conditioned near the no-arbitrage boundary, so its w -space violation rate is an upper bound rather than a real measurement, and we impose its calendar ordering only where consecutive maturities share strike support. The maturity-holdout experiment covers one band and one market. The high F6 rejection rate, the repo-rate proxy, and the single-market sample further limit how far this generalizes. The forward construction, by contrast, is no longer a live concern: a multi-strike regression forward moves the derived panel's implied volatilities by a mean absolute 0.121 volatility points, and the F6 rejection rate by 0.15 percentage points.

None of these limitations touch the paper's central results. The soft-penalty failure mode is diagnosed mechanistically and reproduces across seeds. The classical-baseline violation rates are measured under one shared protocol, applied identically to every method, neural and classical alike. The maturity-holdout test is a controlled, blind comparison in which every baseline gets its best available interpolation rule before we judge the network against it. On this panel, a correctly constrained classical method already delivers an arbitrage-free surface at competitive accuracy, and the network's one surviving structural advantage, being queryable at any maturity by construction, does not turn into a measurable accuracy gain. That is a clear, actionable answer, not an inconclusive one.

DATA AVAILABILITY STATEMENT

The NIFTY 50 index option data used in this study originates from NSE daily F&O bhavcopy files, a historical market-data product available for purchase/download from the National Stock Exchange of India (<https://www.nseindia.com>); the author is not able to redistribute the raw bhavcopy files under NSE's data terms. The derived, de-identified frozen evaluation panel (implied volatilities, log-moneyness, maturities; no raw order-level data), the walk-forward fold definitions, and all code used to build the panel, fit every baseline, train the neural network, and produce every table and figure in this paper will be archived with a versioned DOI before submission; the DOI will be inserted here in the final submission package.

ABBREVIATIONS

IV: implied volatility; NSE: National Stock Exchange (India); F&O: futures and options; SVI: stochastic volatility inspired (parameterization); SSVI: surface SVI; RBF: radial basis function; PINN: physics-informed neural network; MLP: multi-layer perceptron; ATM: at-the-money; OTM: out-of-the-money; RMSE: root-mean-squared error; MAE: mean absolute error; DM: Diebold–Mariano (test); PCP: put–call parity; OI: open interest; UDiFF: NSE’s unified data interchange file format; SLSQP: sequential least-squares quadratic programming.

APPENDIX. PROOF OF CALENDAR MONOTONICITY FOR THE V3/V4 CONSTRUCTION

Section IV-C states the calendar-monotonicity argument around Eq. 5 informally, as a paragraph. We restate it as a proposition and proof.

Proposition 1 (Calendar monotonicity of the v3 parametrization). *Let*

$$w(k, \tau) = \text{softplus}(b(k)) + \sum_{i=1}^3 \text{softplus}(c_i(k)) \varphi_i(\tau),$$

where $\varphi_1(\tau) = \tau$, $\varphi_2(\tau) = \sqrt{\tau}$, $\varphi_3(\tau) = \log(1 + \tau)$, and $b, c_1, c_2, c_3 : \mathbb{R} \rightarrow \mathbb{R}$ are arbitrary measurable functions of k alone, in particular the outputs of a feed-forward network taking only k as input, for any weights. Then

$$\frac{\partial w}{\partial \tau}(k, \tau) \geq 0 \quad \text{for every } k \in \mathbb{R}, \tau > 0.$$

Proof. Each φ_i is differentiable and non-decreasing on $(0, \infty)$:

$$\varphi_1'(\tau) = 1 \geq 0, \quad \varphi_2'(\tau) = \frac{1}{2\sqrt{\tau}} \geq 0, \quad \varphi_3'(\tau) = \frac{1}{1+\tau} \geq 0 \quad (\tau > 0).$$

The softplus function $\text{softplus}(x) = \log(1 + e^x)$ has range $(0, \infty)$ for every $x \in \mathbb{R}$, so $\text{softplus}(c_i(k)) \geq 0$ for every k , and this coefficient does not depend on τ . Differentiating w term by term in τ (the base term $\text{softplus}(b(k))$ is constant in τ and contributes 0),

$$\frac{\partial w}{\partial \tau} = \sum_{i=1}^3 \text{softplus}(c_i(k)) \varphi_i'(\tau).$$

This is a finite sum of products of two non-negative factors, hence non-negative, for every (k, τ) with $\tau > 0$. $\square$

Corollary 1. *Since a surface is free of calendar arbitrage if and only if $\partial w / \partial \tau \geq 0$ everywhere (Eq. 2), every member of the v3 function class is calendar-arbitrage-free, for any network weights, trained or untrained. Calendar-arbitrage-freeness is therefore a property of the function class, not an empirical property of a fitted instance, and requires no penalty term, no λ , and no training signal to hold. The same argument applies unchanged to the state-conditioned extension v4 (Eq. 7): s is held fixed across the maturities of a single daily surface, so $b(k, s)$ and $c_i(k, s)$ are, for that surface, again functions of k alone, and the proposition applies verbatim with $b(\cdot) = b(\cdot, s)$ and $c_i(\cdot) = c_i(\cdot, s)$.*

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